Risk Category 1: Preterm Birth

Clinical Description

Preterm birth is defined as the delivery of a live infant before **37 completed weeks of gestation**. It is one of the leading causes of neonatal morbidity and mortality worldwide and is associated with respiratory distress syndrome, feeding difficulties, neurodevelopmental impairment, infections, cerebral palsy, and increased neonatal mortality. Early identification of maternal demographic, obstetric, and clinical risk factors is essential for timely intervention and improved pregnancy outcomes.

Evidence-Based Clinical Feature Mapping

Clinical Parameter	Evidence from Literature	Evidence Strength	Proposed Weight	Clinical Justification	Sources
Previous Child Deaths (D)	Previous neonatal death increased the risk of preterm birth (**aOR = 4.66**).	Very Strong	**5**	Previous neonatal or child death is one of the strongest indicators of adverse obstetric history and significantly increases the likelihood of subsequent preterm birth.	Baig et al. (2025)
Abortions (A)	Women with ≥2 spontaneous abortions had **aOR = 3.08**. Recurrent pregnancy loss increased the odds of preterm	Very Strong	**5**	Recurrent pregnancy loss reflects underlying cervical, anatomical, hormonal or immunological abnormalities that substantially	Baig et al. (2025)

	birth by **3.87 times**.			increase preterm birth risk.	
Hemoglobin (HB)	Maternal anemia increased the risk of preterm birth (**RR = 1.63**) in a meta-analysis.	Strong	**4**	Maternal anemia reduces oxygen delivery to maternal and fetal tissues, increasing the likelihood of preterm birth and adverse neonatal outcomes.	Rahman et al. (2016); WHO; Cleveland Clinic
Gravida (G)	Gravidity ≥3 was an independent predictor (**aOR = 2.19**).	Strong	**4**	Higher gravidity reflects cumulative reproductive burden and independently contributes to preterm birth risk.	Baig et al. (2025)
Systolic Blood Pressure	Hypertensive disorders significantly increase medically indicated and spontaneous preterm birth.	Moderate	**3**	Elevated systolic blood pressure suggests gestational hypertension or preeclampsia, both strongly associated with placental dysfunction and preterm delivery.	ACOG; WHO; Dutta's Textbook of Obstetrics
Diastolic Blood Pressure	Elevated diastolic blood pressure is associated with hypertensive disorders, placental insufficiency and preterm birth.	Moderate	**3**	Persistent elevation of diastolic blood pressure indicates maternal vascular dysfunction contributing to adverse pregnancy outcomes including preterm birth.	ACOG; WHO; Dutta's Textbook of Obstetrics

Para (P)	Higher parity has been significantly associated with preterm birth, although the association is weaker than previous pregnancy loss or gravidity.	Moderate	**2**	Previous deliveries contribute to obstetric history but should be interpreted together with gravidity and previous pregnancy outcomes.	Baig et al. (2025)
Age	Maternal age <20 years and ≥35 years increases the likelihood of preterm birth.	Moderate	**2**	Extremes of maternal age are recognized demographic risk factors but are not sufficient alone to predict preterm birth.	Cavazos-Rehg et al. (2015)

Clinical Parameters Excluded

Clinical Parameter	Reason for Exclusion	Sources
Living Children (L)	Reflects reproductive history but is not consistently reported as an independent predictor of preterm birth.	Baig et al. (2025)
Random Blood Sugar (RBS)	Primarily indicates gestational diabetes mellitus rather than directly predicting preterm birth.	ACOG; Cleveland Clinic
HbA1c	Marker of long-term glycemic control and gestational diabetes rather than spontaneous preterm birth.	ACOG; Cleveland Clinic
Heart Rate	Tachycardia is more indicative of maternal anemia, infection, or physiological stress than an independent predictor of preterm birth.	MEOWS Guideline; Dutta's Textbook of Obstetrics
Respiratory Rate (RR)	Used for detecting maternal deterioration and sepsis rather than predicting spontaneous preterm birth.	MEOWS Guideline

Body Temperature	Fever is primarily associated with maternal infection. Although infection may precipitate preterm labour, temperature is more appropriately mapped to maternal infection/sepsis.	WHO; ACOG; Cleveland Clinic

---

Final Knowledge Base Mapping

Feature	Final Weight	Basis for Weight
Previous Child Deaths (D)	**5**	Very strong independent predictor (aOR 4.66)
Abortions (A)	**5**	Very strong independent predictor (aOR 3.08–3.87)
Hemoglobin (HB)	**4**	Strong evidence (RR 1.63)
Gravida (G)	**4**	Strong evidence (aOR 2.19)
Systolic Blood Pressure	**3**	Guideline-supported clinical risk factor
Diastolic Blood Pressure	**3**	Guideline-supported clinical risk factor
Para (P)	**2**	Moderate obstetric risk factor
Age	**2**	Demographic risk factor

References (APA 7th Edition)

1. American College of Obstetricians and Gynecologists. (2023). *Preterm Labor and Birth*.
2. Baig, I., Rastogi, P. K., Deo, S., et al. (2025). *Impact of Obstetric History on Preterm Birth: An Observational Study at a Tertiary Care Hospital in North India*. *Cureus, 17*(10), e94558. https://doi.org/10.7759/cureus.94558
3. Cavazos-Rehg, P. A., Krauss, M. J., Spitznagel, E. L., et al. (2015). *Maternal age and risk of labor and delivery complications*. *Maternal and Child Health Journal, 19*(6), 1202–1211.
4. Rahman, M. M., Abe, S. K., Rahman, M. S., et al. (2016). *Maternal anemia and risk of adverse birth and health outcomes in low- and middle-income countries: Systematic review and meta-analysis*. *The American Journal of Clinical Nutrition, 103*(2), 495–504.
5. World Health Organization. (2023). *Preterm Birth*.

6. Cleveland Clinic. (2024). *Preterm Birth: Premature, Causes, Symptoms & Complications*.

Risk Category 2: Gestational Diabetes Mellitus (GDM)

Clinical Overview

Gestational Diabetes Mellitus (GDM) is glucose intolerance that is first recognized during pregnancy. It develops because placental hormones progressively increase insulin resistance, making maternal insulin insufficient to maintain normal blood glucose levels. Although GDM usually resolves after delivery, it substantially increases the mother's future risk of Type 2 diabetes and contributes to adverse maternal and neonatal outcomes if left untreated.

Clinical Feature Weight Assignment

Clinical Feature	Weight	Clinical Justification	Supporting Sources
Random Blood Sugar (RBS)	**5**	Elevated maternal blood glucose is the principal biochemical indicator of gestational hyperglycemia. Persistent hyperglycemia is directly associated with fetal hyperinsulinemia, macrosomia, neonatal hypoglycemia, preterm birth, respiratory distress, and increased maternal complications.	ADA Standards of Care 2026; WHO Guideline; Clinical Parameters Review
HbA1c	**5**	HbA1c reflects long-term glycemic control. Elevated HbA1c before and during pregnancy is associated with congenital anomalies, preeclampsia, macrosomia, and adverse pregnancy outcomes. ADA recommends maintaining HbA1c as close to normal as safely possible before and during pregnancy.	ADA Standards of Care 2026

Age	**2**	Advanced maternal age increases insulin resistance and is a recognized epidemiological risk factor for GDM. However, age alone is insufficient for diagnosis and should be considered a supporting risk factor rather than a primary determinant.	Cleveland Clinic; Clinical Parameters Review

---

Clinical Interpretation

Random Blood Sugar (Weight = 5)

Random Blood Sugar is the strongest predictor available in the current model for identifying gestational hyperglycemia. Elevated glucose levels during pregnancy indicate impaired maternal glucose regulation caused by placental hormone-induced insulin resistance.

Persistent maternal hyperglycemia increases the risk of:

- Fetal macrosomia
- Shoulder dystocia
- Neonatal hypoglycemia
- Respiratory distress syndrome
- Preterm birth
- Cesarean delivery

Because RBS directly reflects current maternal glycemic status, it is assigned the highest evidence weight.

Evidence Sources

- ADA Standards of Care 2026
- WHO Guideline
- Clinical Parameters Review

HbA1c (Weight = 5)

HbA1c reflects the average blood glucose level over approximately three months and provides evidence of chronic hyperglycemia.

Elevated HbA1c has been associated with:

- Congenital malformations
- Preeclampsia
- Macrosomia
- Preterm birth

- Poor neonatal outcomes

Because HbA1c complements instantaneous glucose measurements by capturing long-term glycemic exposure, it receives the same highest evidence weight as RBS.

Evidence Sources

- ADA Standards of Care 2026

Maternal Age (Weight = 2)

Advanced maternal age is associated with increasing insulin resistance and a higher incidence of gestational diabetes.

However,

- Age is not diagnostic.
- Many younger women also develop GDM.
- Age mainly modifies baseline risk.

Therefore, maternal age functions as a supporting epidemiological factor and is assigned a lower weight.

Evidence Sources

- Cleveland Clinic
- Clinical Parameters Review

Final Evidence-Based Feature Weights

Feature	Final Weight	Supporting Sources
Random Blood Sugar (RBS)	**5**	ADA 2026; WHO Guideline; Clinical Parameters Review
HbA1c	**5**	ADA 2026
Maternal Age	**2**	Cleveland Clinic; Clinical Parameters Review

---

Final Knowledge Base Entry

"Gestational Diabetes Mellitus (GDM)": {

```
  "features": {
    "RBS": 5,
    "HBA1C": 5,
    "Age": 2
  }
}
```


Why these weights?

Unlike the earlier version where the weights were based primarily on heuristic reasoning, these values are now explicitly justified by published clinical guidance and evidence:

- **RBS (5)** because it is the primary indicator of current gestational hyperglycemia and is directly linked to maternal and fetal complications.
- **HbA1c (5)** because it captures chronic glycemic exposure and predicts adverse pregnancy outcomes.
- **Age (2)** because it is an epidemiological risk modifier rather than a diagnostic biomarker.

Risk Category 3: Fetal Growth Restriction (FGR)

Clinical Overview

Fetal Growth Restriction (FGR), also referred to as Intrauterine Growth Restriction (IUGR), is a condition in which the fetus fails to achieve its genetically determined growth potential. Clinically, FGR is defined as an estimated fetal weight below the 10th percentile for gestational age. Unlike constitutionally small but healthy fetuses, FGR results from pathological processes that reduce oxygen and nutrient delivery to the fetus, most commonly because of placental insufficiency. FGR is associated with increased risks of stillbirth, neonatal morbidity, preterm birth, neurodevelopmental impairment, and long-term cardiovascular and metabolic disease.

Clinical Feature Weight Assignment

Clinical Feature	Weight	Clinical Justification	Supporting Sources
Hemoglobin (HB)	**5**	Maternal anemia reduces oxygen-carrying capacity, leading to chronic fetal hypoxia, impaired placental oxygen delivery, low birth weight, and fetal growth restriction. It is one of the strongest directly measurable maternal predictors available in the model.	Clinical Parameters Review; WHO
Systolic Blood Pressure (SBP)	**4**	Maternal hypertension causes placental vascular dysfunction, reducing uteroplacental blood flow and fetal nutrient supply. It is a major cause of placental insufficiency and FGR.	Clinical Parameters Review; Cleveland Clinic
Diastolic Blood Pressure (DBP)	**4**	Elevated DBP reflects hypertensive disorders that impair placental perfusion and increase the likelihood of fetal growth restriction, placental abruption, and preterm birth.	Clinical Parameters Review
Age	**2**	Advanced maternal age increases the risk of placental dysfunction and poor fetal growth, although age alone cannot diagnose FGR and therefore acts as a supporting epidemiological factor.	Clinical Parameters Review; Cleveland Clinic

---

Clinical Interpretation

Hemoglobin (Weight = 5)

Maternal hemoglobin concentration determines the blood's oxygen-carrying capacity. Low hemoglobin levels reduce oxygen delivery to the placenta and fetus, resulting in chronic fetal hypoxia and impaired fetal growth.

Maternal anemia has been associated with:

- Fetal Growth Restriction
- Low Birth Weight
- Preterm Birth
- Perinatal mortality
- Neonatal mortality

Among all laboratory variables available in the current prediction model, hemoglobin has the strongest direct physiological relationship with fetal oxygen supply, making it the highest-weight feature for this risk category.

Evidence Sources

- Clinical Parameters Review
- WHO Maternal Nutrition & FGR Review

Systolic Blood Pressure (Weight = 4)

Elevated systolic blood pressure is an indicator of gestational hypertension and preeclampsia.

Hypertensive disorders cause:

- Placental insufficiency
- Reduced uteroplacental perfusion
- Chronic fetal hypoxia
- Impaired nutrient delivery

These mechanisms significantly increase the likelihood of fetal growth restriction.

Since blood pressure reflects one of the principal pathological mechanisms responsible for FGR, SBP receives a high evidence weight.

Evidence Sources

- Clinical Parameters Review
- Cleveland Clinic FGR Review

Diastolic Blood Pressure (Weight = 4)

Elevated diastolic blood pressure indicates sustained vascular resistance and worsening placental perfusion.

High DBP has been linked with:

- Placental insufficiency
- Fetal Growth Restriction
- Placental abruption
- Preterm birth

DBP complements systolic pressure in identifying hypertensive disease severity, therefore receiving the same evidence weight.

Evidence Sources

- Clinical Parameters Review

Maternal Age (Weight = 2)

Advanced maternal age is associated with an increased incidence of placental dysfunction and fetal growth abnormalities.

However,

- many younger women also develop FGR,
- age alone cannot establish the diagnosis,
- it mainly modifies baseline population risk.

Consequently, maternal age functions as a supporting epidemiological variable.

Evidence Sources

- Clinical Parameters Review
- Cleveland Clinic

Why Other Model Features Were Not Included

Although your model contains additional obstetric variables (Gravida, Para, Living Children, Abortions, Child Deaths, RBS, HbA1c, Heart Rate, Respiratory Rate, Body Temperature), current high-quality evidence does not support them as primary physiological predictors of FGR:

- **Gravida, Para, Living Children, Abortions, Child Deaths** describe reproductive history but are not direct indicators of impaired fetal growth.
- **Random Blood Sugar (RBS)** and **HbA1c** primarily predict gestational diabetes. Poorly controlled diabetes more commonly causes **macrosomia** rather than classical FGR, except in advanced diabetic vasculopathy.
- **Heart Rate**, **Respiratory Rate**, and **Body Temperature** reflect maternal systemic illness and are useful for general maternal assessment but have limited direct evidence linking them to FGR prediction.

Therefore, they are excluded from the FGR knowledge base to maintain clinical specificity.

Final Evidence-Based Feature Weights

Feature	Final Weight	Supporting Sources
Hemoglobin (HB)	**5**	WHO; Clinical Parameters Review
Systolic BP	**4**	Cleveland Clinic; Clinical Parameters Review
Diastolic BP	**4**	Clinical Parameters Review
Age	**2**	Cleveland Clinic; Clinical Parameters Review

Final Knowledge Base Entry


```
"Fetal Growth Restriction (FGR)": {
  "features": {
    "HB": 5,
    "SystolicBP": 4,
    "DiastolicBP": 4,
    "Age": 2
  }
}
```


Why these weights?

The assigned weights are based on the strength of the biological relationship between each model feature and fetal growth restriction:

- **Hemoglobin (5):** Direct determinant of fetal oxygen delivery; maternal anemia is strongly associated with FGR, low birth weight, and adverse perinatal outcomes.
- **Systolic BP (4):** Reflects hypertensive disorders that reduce uteroplacental blood flow and contribute to placental insufficiency.
- **Diastolic BP (4):** Complements SBP in identifying placental vascular disease and impaired fetal perfusion.
- **Age (2):** Acts as an epidemiological modifier of FGR risk but is not independently diagnostic.

Risk Category 4: Maternal Anemia

Clinical Overview

Maternal anemia is one of the most common medical disorders during pregnancy and is defined by the World Health Organization (WHO) as a hemoglobin concentration of **<11 g/dL** during the first and third trimesters (or **<10.5 g/dL** during the second trimester because of physiological hemodilution). Iron deficiency is the most frequent cause, although folate deficiency, vitamin B12 deficiency, infections, and inherited hematologic disorders may also

contribute. Maternal anemia reduces the oxygen-carrying capacity of the blood, compromising oxygen delivery to maternal tissues and the developing fetus. Severe or untreated anemia increases the risk of maternal morbidity, postpartum hemorrhage, fetal growth restriction, preterm birth, low birth weight, and perinatal mortality.

Clinical Feature Weight Assignment

Clinical Feature	Weight	Clinical Justification	Supporting Sources
Hemoglobin (HB)	**5**	Hemoglobin is the diagnostic biomarker for maternal anemia. Reduced hemoglobin directly reflects diminished oxygen-carrying capacity and is the primary laboratory criterion used worldwide to diagnose and classify anemia severity during pregnancy.	WHO; Clinical Parameters Review; Cleveland Clinic
Heart Rate	**2**	Moderate to severe anemia often produces compensatory tachycardia due to reduced oxygen delivery. Heart rate reflects physiological compensation rather than the disease itself, making it a supportive feature.	Cleveland Clinic; MEOWS guideline in Clinical Parameters Review
Respiratory Rate	**2**	Increased respiratory rate may occur in significant anemia as the body attempts to improve oxygen delivery. RR is a secondary physiological response and not diagnostic.	Clinical Parameters Review

---

Clinical Interpretation

Hemoglobin (Weight = 5)

Hemoglobin is the principal oxygen-carrying protein within red blood cells and is the gold-standard laboratory measurement used to diagnose maternal anemia.

Reduced hemoglobin causes:

- Decreased oxygen delivery to maternal tissues
- Reduced fetal oxygenation
- Maternal fatigue and weakness
- Increased risk of fetal growth restriction
- Increased risk of low birth weight

- Increased risk of preterm birth
- Increased perinatal mortality

WHO defines anemia during pregnancy as:

- **Hb <11 g/dL** (1st and 3rd trimester)
- **Hb <10.5 g/dL** (2nd trimester)

Because hemoglobin directly determines both diagnosis and severity of maternal anemia, it receives the highest evidence weight.

Evidence Sources

- WHO Prevention and Management of Severe Anaemia in Pregnancy
- Clinical Parameters Review
- Cleveland Clinic

Heart Rate (Weight = 2)

When hemoglobin falls, the body compensates by increasing cardiac output to maintain oxygen delivery to tissues.

Patients with significant anemia frequently develop:

- Tachycardia
- Palpitations
- Exercise intolerance

Heart rate therefore reflects the body's physiological response to anemia rather than anemia itself.

It provides supportive clinical evidence but cannot diagnose maternal anemia independently.

Evidence Sources

- Cleveland Clinic
- Clinical Parameters Review

Respiratory Rate (Weight = 2)

Reduced oxygen transport caused by anemia may stimulate an increase in respiratory rate to improve oxygen uptake.

Elevated respiratory rate is therefore associated with:

- Moderate anemia

- Severe anemia
- Maternal physiological compensation

Since respiratory rate is a non-specific response that occurs in many maternal illnesses, it receives a supportive weight.

Evidence Sources

- Clinical Parameters Review

Why Other Model Features Were Not Included

Although several variables exist in the prediction model, current clinical evidence does not support them as primary predictors of maternal anemia:

- **Age** influences anemia prevalence in some populations but is not diagnostic.
- **Gravida, Para, Living Children, Abortions, Child Deaths** describe obstetric history and may influence nutritional status over time but are not reliable physiological indicators of current anemia.
- **Systolic BP** and **Diastolic BP** are mainly markers of hypertensive disorders and are not routinely altered by uncomplicated anemia.
- **Random Blood Sugar (RBS)** and **HbA1c** assess glucose metabolism rather than oxygen-carrying capacity.
- **Body Temperature** reflects infection and inflammation rather than anemia.

Therefore, these features are excluded to preserve the specificity of the maternal anemia knowledge base.

Final Evidence-Based Feature Weights

Feature	Final Weight	Supporting Sources
Hemoglobin (HB)	**5**	WHO; Clinical Parameters Review; Cleveland Clinic
Heart Rate	**2**	Cleveland Clinic; Clinical Parameters Review
Respiratory Rate	**2**	Clinical Parameters Review

Final Knowledge Base Entry


```
"Maternal Anemia": {
  "features": {
     "HB": 5,
     "HeartRate": 2,
     "RR": 2
  }
}
```


Why these weights?

The weights reflect the strength of association between each available model feature and maternal anemia:

- **Hemoglobin (5):** Direct diagnostic biomarker and the single most important indicator of anemia severity.
- **Heart Rate (2):** Represents compensatory cardiovascular adaptation to reduced oxygen-carrying capacity but is non-specific.
- **Respiratory Rate (2):** Reflects compensatory respiratory response in moderate to severe anemia but is also non-specific.

Risk Category 5: Maternal Infection

Clinical Overview

Maternal infection during pregnancy includes bacterial, viral, fungal, and parasitic infections that may affect both maternal and fetal health. Pregnancy induces physiological and immunological adaptations that increase susceptibility to infections while maintaining tolerance to the fetus. Depending on the pathogen and gestational age, maternal infections may lead to miscarriage, preterm labor, premature rupture of membranes, fetal growth restriction, congenital infection, stillbirth, neonatal sepsis, and maternal morbidity. Early recognition through maternal vital signs and prompt treatment is therefore essential.

Clinical Feature Weight Assignment

Clinical Feature	Weight	Clinical Justification	Supporting Sources
Body Temperature	**5**	Fever is the primary physiological indicator of maternal infection and is routinely used for infection screening during pregnancy.	Clinical Parameters Review; Frontiers Review
Heart Rate	**3**	Infection commonly produces maternal tachycardia through systemic inflammatory response and is included in obstetric early warning systems.	Clinical Parameters Review; Frontiers Review
Respiratory Rate	**3**	Increased respiratory rate is an important physiological response to systemic infection and sepsis and is used in maternal deterioration assessment.	Clinical Parameters Review

---

Clinical Interpretation

Body Temperature (Weight = 5)

Body temperature is the strongest available indicator of maternal infection in your model.

Maternal fever may indicate:

- Intra-amniotic infection (chorioamnionitis)
- Urinary tract infection
- Viral infection
- Bacterial infection
- Sepsis

The clinical parameters document identifies:

- Normal: **36.5–37.5°C**
- Fever: **≥38°C**

Maternal fever has been associated with:

- Preterm labor
- Fetal tachycardia
- Neonatal sepsis
- Maternal morbidity

Since fever is the primary clinical manifestation of infection and the only direct infection-related variable in your dataset, it receives the highest weight.

Evidence Sources

- Clinical Parameters Review
- *Infections and Pregnancy: Effects on Maternal and Child Health*

Heart Rate (Weight = 3)

Maternal infection activates the systemic inflammatory response, often causing **tachycardia**.

Elevated heart rate is commonly observed in:

- Sepsis
- Chorioamnionitis
- Systemic bacterial infections
- Viral illnesses with fever

Heart rate is also part of the **Modified Early Obstetric Warning System (MEOWS)** used to identify early maternal deterioration. It is an important supportive feature but is not specific for infection.

Evidence Sources

- Clinical Parameters Review
- Frontiers Review

Respiratory Rate (Weight = 3)

Systemic infection may increase respiratory rate through fever, metabolic stress, or early sepsis.

An elevated respiratory rate is commonly seen in:

- Severe maternal infection
- Sepsis
- Pneumonia
- Systemic inflammatory response

Because respiratory rate is a recognized marker of maternal deterioration but is not infection-specific, it receives a moderate weight.

Evidence Sources

- Clinical Parameters Review

Why Other Model Features Were Not Included

Although the model contains additional features, current evidence does not support them as reliable indicators of maternal infection:

- **Age** may influence susceptibility to infection but is not a clinical marker of active infection.
- **Gravida, Para, Living Children, Abortions, Child Deaths** describe obstetric history rather than current infectious status.
- **Systolic BP** and **Diastolic BP** may decrease in septic shock or vary with illness severity but are not early or specific indicators of infection.
- **Hemoglobin (HB)** reflects anemia rather than infection.
- **Random Blood Sugar (RBS)** and **HbA1c** assess glucose metabolism; diabetes increases infection risk but these values do not indicate an active infection.

Therefore, these variables are excluded to maintain specificity.

Final Evidence-Based Feature Weights

Feature	Final Weight	Supporting Sources
Body Temperature	**5**	Clinical Parameters Review; Frontiers Review
Heart Rate	**3**	Clinical Parameters Review; Frontiers Review
Respiratory Rate	**3**	Clinical Parameters Review

Final Knowledge Base Entry

```
"Maternal Infection": {
  "features": {
     "BodyTemp": 5,
     "HeartRate": 3,
     "RR": 3
  }
}
```

Why these weights?

- **Body Temperature (5):** Direct clinical indicator of infection and the strongest infection-related feature available in your model.
- **Heart Rate (3):** Important physiological marker of systemic inflammatory response and maternal deterioration.
- **Respiratory Rate (3):** Reflects respiratory compensation and early sepsis but is less specific than fever.

Risk Category 6: Hypertensive Disorders of Pregnancy

Clinical Overview

Hypertensive disorders of pregnancy (HDP) include gestational hypertension, preeclampsia, eclampsia, chronic hypertension, and chronic hypertension with superimposed preeclampsia. They are among the leading causes of maternal and perinatal morbidity and mortality worldwide. HDP can impair placental perfusion, leading to fetal growth restriction, placental abruption, preterm birth, stillbirth, maternal organ dysfunction, stroke, and eclampsia. Early identification through routine blood pressure monitoring is therefore a cornerstone of antenatal care.

Clinical Feature Weight Assignment

Clinical Feature	Weight	Clinical Justification	Supporting Sources
Systolic BP	**5**	Primary diagnostic criterion for gestational hypertension and preeclampsia. Higher values correlate with disease severity.	ACOG; Clinical Parameters Review
Diastolic BP	**5**	Equally important diagnostic criterion. DBP ≥90 mmHg defines hypertension and ≥110 mmHg indicates severe disease.	ACOG; Clinical Parameters Review

Age	**2**	Advanced maternal age (≥35 years) is a recognized risk factor for preeclampsia, while very young mothers also have increased risk.	ACOG; Clinical Parameters Review
Heart Rate	**2**	Tachycardia is not diagnostic but may accompany severe hypertensive disease and maternal deterioration.	Clinical Parameters Review

---

Clinical Interpretation

Systolic Blood Pressure (Weight = 5)

Systolic blood pressure is the single most important clinical parameter for detecting hypertensive disorders during pregnancy.

According to ACOG:

- **≥140 mmHg** → Gestational hypertension
- **≥160 mmHg** → Severe hypertension requiring urgent management

Elevated SBP is associated with:

- Preeclampsia
- Eclampsia
- Placental abruption
- Fetal growth restriction
- Preterm birth
- Maternal stroke
- Maternal organ dysfunction

Because SBP directly defines disease severity, it receives the highest weight.

Evidence Sources

- ACOG Practice Bulletin
- Clinical Parameters Review

Diastolic Blood Pressure (Weight = 5)

Diastolic blood pressure is equally important for diagnosing hypertensive disorders.

Clinical thresholds:

- **≥90 mmHg** → Gestational hypertension

- **≥110 mmHg** → Severe hypertension

Persistent elevation reflects increased vascular resistance and placental dysfunction and is associated with:

- Preeclampsia
- Eclampsia
- Placental insufficiency
- Fetal growth restriction
- Preterm birth

Since international diagnostic criteria require assessment of both systolic and diastolic pressure, DBP also receives the highest weight.

Evidence Sources

- ACOG Practice Bulletin
- Clinical Parameters Review

Age (Weight = 2)

Maternal age is a recognized non-modifiable risk factor for hypertensive disorders.

Higher risk is observed in:

- Advanced maternal age (≥35 years)
- Very young mothers (≤19 years)

However, age contributes to **predisposition** rather than diagnosis, so it receives a lower weight.

Evidence Sources

- ACOG Risk Factors
- Clinical Parameters Review

Heart Rate (Weight = 2)

Heart rate is not part of the diagnostic criteria for hypertensive disorders but may increase with severe disease, pain, stress, or maternal deterioration. It is included in obstetric early warning systems as a supportive physiological indicator.

Because tachycardia is nonspecific, it receives a low supporting weight.

Evidence Source

- Clinical Parameters Review

Why Other Model Features Were Not Included

The following model features are not direct indicators of hypertensive disorders:

- **Gravida, Para, Living Children, Abortions, Child Deaths:** Obstetric history variables that are not used to diagnose HDP.
- **Random Blood Sugar and HbA1c:** Diabetes increases the risk of preeclampsia, but these measurements do not indicate active hypertensive disease.
- **Hemoglobin:** May change in severe disease due to hemoconcentration or HELLP syndrome, but is not a diagnostic marker.
- **Body Temperature:** Fever is unrelated to hypertensive disorders unless another condition is present.
- **Respiratory Rate:** May increase in pulmonary edema or severe disease but is a late, nonspecific finding.

Final Evidence-Based Feature Weights

Feature	Final Weight
SystolicBP	**5**
DiastolicBP	**5**
Age	**2**
HeartRate	**2**

Final Knowledge Base Entry

```
"Hypertensive Disorders of Pregnancy": {
  "features": {
    "SystolicBP": 5,
    "DiastolicBP": 5,
    "Age": 2,
    "HeartRate": 2
```

```
  }
}
```